Predicting the Next Category and Region of a Visit

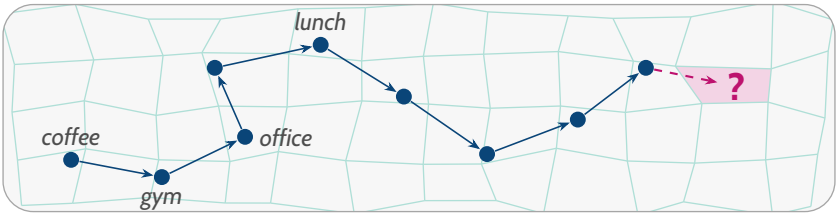
A Check-in-Level Multi-Task Study on Mobility Data

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MobiWac 2026 · Paris · October 26, 2026

One person's day, and two questions



What kind of place?

Next category: one of 7 types, such as food, shopping, nightlife. → *what the user will want*

Which neighborhood?

Next region: its census tract (Istanbul: *mahalle*), 520 to 8,501. → *where to prepare*

Check-in: a visit shared on a location-based social network such as Gowalla: a place (a *point of interest*, POI) and a time (Cho 2011). Uses: caching, capacity, demand anticipation (Bastug 2014; Moura 2025). A tract is not a radio cell: we claim nothing about cells or handover.

Two models, or one?

- **Two dedicated models**: one per question; two things to train, version and serve.
- **One joint model**: one artifact to deploy, and the tasks might help each other. But shared parameters can settle on a compromise good for neither task (Caruana 1997).

Multi-task models in this field use category and region as helpers for next-place prediction (MCMG, Sun 2024; HMT-GRN, Lim 2022). To our knowledge, fine-grained region as an end target of equal standing is underexplored.

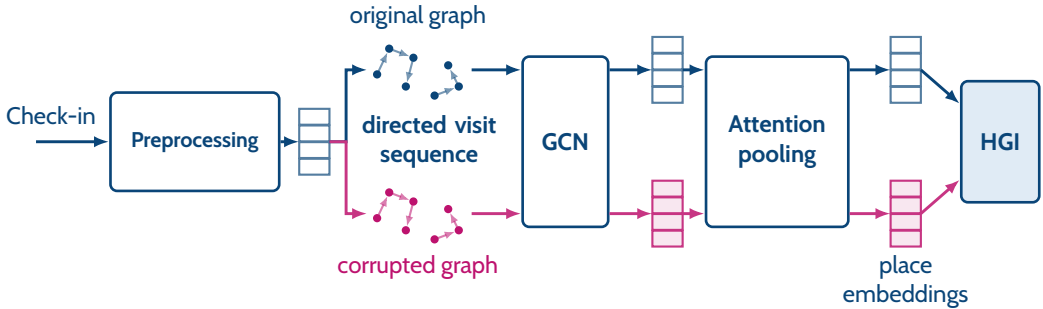
The question

Can one model predict both the category and the region of a user's next visit, and what does it cost?

Short answer: yes, at a measured price.

Check2HGI: a fourth level below the place

Place embeddings: one vector per place, DGI (Veličković 2019), HGI (Huang 2023). Two visits to one café look identical.



- **Each visit carries** its place's category, the hour and day of week, and four elapsed-time values → one **64-d** vector per visit, **the vector our model reads**; pooled per place, it feeds the levels above.
- Trained **without the task labels**, and **built from the past alone**: edges between consecutive visits run forward in time only.

How it was measured

Regions	census tracts (Istanbul: *mahalle*), **520** to **8,501**
Split	by user: the prediction models never train on a test user's visits
Scored	category: **macro-F1** (always-majority: 5.7 to 7.3)
	region: **Acc@10**, is the true region in the top ten? (random: at most about 2%)
Repeated	**20 trained models** per configuration (4 seeds $\times$ 5 folds)
Decided	for region, "not worse" $=$ within **2 Acc@10 points**, written down before any result was seen (TOST, Lakens 2017)

Result 1: the representation, at every dataset

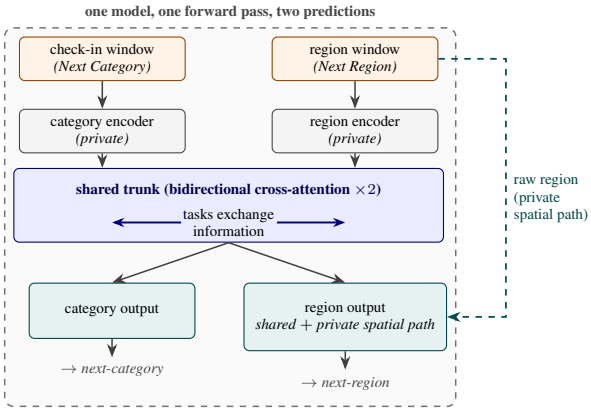
Next category, single-task model, seed 0 — only the input changes

Dataset	Check-in level	Place level (HGI)	Δ
Alabama	**30.77** $\pm$1.16	29.15 $\pm$0.84	$+1.62$
Arizona	**34.51** $\pm$1.15	31.93 $\pm$1.08	$+2.58$
Istanbul	**35.35** $\pm$0.89	32.54 $\pm$1.03	$+2.81$
Florida	**37.36** $\pm$0.42	37.13 $\pm$0.41	$+0.23$
California	**35.62** $\pm$0.47	34.74 $\pm$0.41	$+0.88$
Texas	**36.32** $\pm$0.51	35.33 $\pm$0.48	$+0.99$

macro-F1; five matched folds, $\pm$ is the fold sd.

Larger at the three smaller datasets ($+1.6$ to $+2.8$), under one point at the three largest. A paired test separates the two at five of six; not at Florida.

One model for two tasks



Multi-task learning: one model does several jobs at once by sharing most of its parts.

The risk: what the tasks share can settle on a compromise worse for each.

Here: each task has its own input and encoder. In the shared trunk, attention lets each task read the other's features while keeping its own weights: **they share by exchanging information**.

A private spatial path keeps what is purely spatial out of the exchange.

One model, two answers: the price

	Next region (Acc@10)			**Next category** (macro-F1)		
	Ded.	Joint	Δ	Ded.	Joint	Δ
Alabama	**70.12**	69.24	−0.87	**30.77**	30.59	−0.19
Arizona	**59.48**	59.04	−0.44	**34.57**	**34.57**	−0.00
Istanbul	**75.16**	75.08	−0.08	35.34	**35.42**	+0.08
Florida	**76.69**	76.54	−0.16	37.35	**37.55**	+0.19
California	63.48	**64.54**	+1.06	**35.63**	**35.63**	−0.00
Texas	64.94	**66.15**	+1.21	**36.33**	36.19	−0.13

Ded.: two dedicated models; Joint: one model. **Green**: outperforms (Holm-corrected). Bold: the larger. **Region**: the other four are deficits, not ties, inside the 2-point margin. **Category**: every interval within half a point; only Florida survives correction. Δ: paired mean over four seeds (can differ from the rounded columns).

The trade is a measured one and not a free substitution.

And against published systems

Next category (macro-F1)

	POI-RGNN	Ded.	Joint
AL	23.80	**30.77**	30.59
AZ	27.64	**34.57**	**34.57**
IST	30.12	35.34	**35.42**
FL	33.35	37.35	**37.55**
CA	30.71	**35.63**	**35.63**
TX	32.08	**36.33**	36.19

Next region (Acc@10)

	HMT-GRN	ReHDM	STAN	Ded.	Joint
AL	63.19	65.38	60.72	**70.12**	69.24
AZ	52.11	53.00	49.86	**59.48**	59.04
IST	68.06	69.33	61.86	**75.16**	75.08
FL	72.76	64.49	72.99	**76.69**	76.54
CA	59.59	50.26	58.61	63.48	**64.54**
TX	61.39	48.81	61.75	64.94	**66.15**

Both our models are above every system we compared against; the joint model by at least **4.11** (category) and **3.55** (region). That margin includes the representation's advantage, so it is not evidence for sharing. Bold: best in row; underline: second.

POI-RGNN (Capanema 2023): next category, re-implemented. **HMT-GRN** (Lim 2022): multi-task next place + regions, next-place parts removed; our windows and folds, best epoch; not the full published system. **STAN** (Luo 2021): re-implemented, its own representations and splits, ranks regions. **ReHDM** (Li 2025): its own split.

What a service would receive

- **Time to act** — half of the next visits come within **0.4 h** (Florida) to **5.5 h** (Istanbul) of the previous one; 5 to 27% come more than 3 days later.
- **A shortlist** — ask the model for its ten most likely neighborhoods. In California (8,501) the true one is among them **64.54%** of the time; in Texas (6,553), **66.15%**. This is the Acc@10 of slide 7, read as a hit rate.
- **For scale** — a transition table with no learning reaches 51 to 72 Acc@10; the joint model is 4 to 10 points above it everywhere.

We built and evaluated no such service: this is motivation, not a measured service result.

What it costs, and four limits

THE COST

Parameters, joint vs **the two dedicated models combined**:

AL 4.2 M vs 1.9 M
CA 5.2 M vs 2.8 M

One pass costs **more** compute than the two small models together. The gain is operational: one artifact to train, version and deploy; two answers.

FOUR LIMITS

1. Representation trained **once over all users**; a per-fold rebuild moved the dedicated models by at most 0.17 macro-F1 and 0.13 Acc@10 (one seed, the datasets checked).
2. Epoch selection reads the scored fold: **absolute scores are optimistic**; joint vs dedicated is affected far less.
3. Each visit's vector uses **only the visits before it**: no later context enters, which is also what keeps a vector from carrying its target (slide 3).
4. The joint model is **larger**, so its region gains at Texas and California may come from size, not sharing. The one-model result does not depend on which.

Conclusion

- **Built**: a check-in-level representation that extends hierarchical graph infomax from the place to the individual visit; and one model that answers both questions.
- **Measured**: on region, the joint model outperforms the dedicated model at Texas and California and stays within the 2-point margin elsewhere (deficits, not ties); on category, within half a point. A measured trade, not a free substitution.
- **Not shown**: that sharing is the cause. On category the representation matters more than one model versus two; the region gains may come from size.
- **Next**: a joint-model search at Texas and California; separating size from sharing; how far a miss lands, inside a service that uses the shortlist.

What a user will do next, and where, from one model instead of two.

Thank you

Questions?

Code (model, representation, baselines, statistical tests)
`github.com/VitorHugoOli/PoiMtlNet/tree/mobiwac`

Data (public) · category-annotated Gowalla · Massive-STEPS (Istanbul)

Supported by **FAPEMIG**, **CNPq** and **CAPES**.

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B1 · Why census tracts

- Predicting over a partition of the map is standard, usually with a **grid cell**; we use **official neighborhood-scale units**;
- Next region is classification over all the dataset's regions:

	Istanbul	AL	AZ	FL	TX	CA
Regions	520	1,109	1,547	4,703	6,553	8,501

B2 · The margin, and the tests

- **Registered before any result was read**: superiority for category, non-inferiority for region with a margin of **2 Acc@10 points**, paired Wilcoxon;
- **Why 2**: a service that acts on which neighborhood will be busy would not behave differently below it; the seed-to-seed sd of the paired difference is 0.02 to 0.16;
- **Reported**: paired t on the four per-seed means, 90% interval, Holm across the six datasets per task; at four seeds the exact Wilcoxon cannot go below 0.0625. Both tests are in the code release;
- Category has **no registered equivalence margin**: a difference that fails superiority is unresolved, never a match;
- The two region gains are **secondary results**, outside the registered plan.

B3 · The Markov floor

- A first-order Markov chain over region transitions, no learning: **51.23 to 72.47** Acc@10;
- The joint model exceeds it by $+4.1$ **to** $+10.0$ at every dataset; the narrowest is **Florida,** $+4.07$.

B4 · Can a visit vector see its target?

- Each visit node draws on the **visits that precede it**: the vector encodes the visit with its own history;
- The graph does **not pass information from a later visit back to an earlier one**, in training or at readout;
- That is what keeps a node from carrying a feature of the target it is used to predict;
- The graph is trained without the task labels: it never sees the next category or the next region.

B5 · One model instead of two

- **4.2 M vs 1.9 M** parameters at Alabama, **5.2 M vs 2.8 M** at California, against **the two dedicated models combined**;
- One pass costs more compute than running the two small models;
- The gain is operational: one artifact to train, version and deploy; one pass returns both answers.

B6 · Then what is multi-task learning for?

- On category, the representation moves the score **more than the choice between one model and two**;
- Whether the exchange between the tasks adds anything beyond the representation is not separated by the evidence here;
- What one model gives for certain is operational: one artifact, two answers.

B7 · The external systems

- **HMT-GRN** (Lim 2022) — primary region comparison: our windows and folds, one seed, best validation epoch; its multi-task structure kept, the graph components and hierarchical beam search removed, a per-fold region-transition prior added (our own reported models do not use it); not the complete published system;
- **STAN** (Luo 2021) — re-implemented with its own representations and splits from the raw check-ins, output adapted to rank regions, seed 0;
- **ReHDM** (Li 2025) — its own published chronological split; single seed at TX and CA;
- **POI-RGNN** (Capanema 2023) — next category, re-implemented, non-overlapping windows.

B8 · More regions, more gain?

	Regions	Region Δ (Acc@10)
Texas	6,553	$+1.21$
California	8,501	$+1.06$

California has more regions than Texas and a **smaller** gain.

An observation, not a law.

B9 · Is it just the raw features?

Control: the place vector + the same raw per-visit features (category one-hot, hour, day of week), same single-task model. Run at three datasets.

	Control gain	Check-in-level gain
Alabama	$+1.73$	$+1.62$
Arizona	$+1.70$	$+2.58$
Florida	$+1.02$	$+0.23$

macro-F1 over the place-level representation, category axis.

"Most of the category difference is therefore already available in the raw per-visit features, and this control does not separate what the check-in-level representation adds beyond them on this axis."

B10 · The intervals

Joint minus dedicated, 90% interval of the paired difference

	Next region (Acc@10)		**Next category** (macro-F1)	
	Δ	90% CI	Δ	90% CI
Alabama	-0.87	$[-1.00, -0.75]$	-0.19	$[-0.33, -0.04]$
Arizona	-0.44	$[-0.62, -0.25]$	-0.00	$[-0.04, +0.03]$
Istanbul	-0.08	$[-0.16, -0.002]$	$+0.08$	$[+0.01, +0.15]$
Florida	-0.16	$[-0.19, -0.13]$	$+0.19$	$[+0.14, +0.25]$
California	$+1.06$	$[+1.03, +1.08]$	-0.00	$[-0.03, +0.02]$
Texas	$+1.21$	$[+1.13, +1.29]$	-0.13	$[-0.19, -0.08]$

Green: outperforms, Holm-corrected — region at Texas ($p = 0.00013$) and California ($p < 10^{-4}$), 20 of 20 folds; category at Florida ($p = 0.011$), 19 of 20.

Every region interval of the other four lies **entirely below zero**: deficits, not ties (Istanbul only just).

B11 · References

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